A multi-RF head receiver array for AI-native sensing and communications

Group 20

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1. Summary

This project aims to develop a flexible multi-band RF sensing platform based on the Integrated Sensing and Communications (ISAC) paradigm, which is widely recognized as a key technology for future 6G networks. In contrast to conventional approaches that employ separate hardware for communication, radar sensing, spectrum monitoring, and target detection, the proposed platform integrates these capabilities into a unified receiver architecture covering the 800 MHz–8 GHz frequency range. By using multiple antenna arrays and advanced signal processing techniques, the system can simultaneously receive communication signals, monitor spectrum activity, detect and track drones, estimate their location, and analyze radar signatures. One of its major applications is counter-drone and surveillance operations. Through AI-assisted micro-Doppler analysis, the system can potentially distinguish drones from birds and classify different drone types based on the unique signatures generated by their rotating propellers. Similar capabilities currently exist in military electronic warfare, spectrum intelligence, and counter-UAP sensing, but they are often implemented using multiple expensive subsystems. What makes this project unique is that it brings together communication, radar sensing, spectrum analysis, antenna array processing, and AI-based target recognition in one flexible platform. This integrated approach provides a practical environment for research and development in future 6G ISAC systems, location and spectrum intelligence and autonomous sensing networks.

2. Description

Since the receiver must be able to grab a wider range of the spectrum it must operate in the frequency band 800-8000 MHz, have a noise figure better than 2 dB, gain of 90 dB, and an instantaneous bandwidth of 400 MHz. The receiver assumes low-phase noise LO inputs from external sources that can be tuned as needed. Also this receiver has four RF input sections for four frequency diverse arrays, operating in the 800-1600 band, and 1000-3000 MHz band, 2000-8000 MHz and a narrowband 5.7-6.0 GHz array. (eg:- 1000-3000 MHz for Mobile Communication ,2000 - 8000 MHz for Spectrum Sensing and 5.7 - 6.0 GHz for Drone Sensing). The first three arrays contain wideband Vivaldi antennas, and the fourth array contains patch antennas. All elements have independent LNAs with about 40 dB gain. The receivers share a common IF gain stage and have RF switches controlled via embedded control to switch band. Tunable filters with controllable center frequency and Q factor are designed into each receiver arm for each band. Any combination of bands can be active at a time. The IF section supports wideband communications, radar and ISAC waveforms, location and spectrum intelligence. The project will entail a full PCB design and implementation of a 4-channel receiver with corresponding antennas and RF systems. Gain, and other parameters can be controlled via USB, ethernet or a similar interface through an embedded processor. .

3. Deliverables

1. **Multi-RF head receiver array**: A four-channel multi-band RF receiver platform capable of operating across the 800 MHz–8 GHz spectrum. The system will integrate multiple antenna arrays, RF front-end circuitry, tunable filtering, and programmable gain control.
2. **Software application**: A software suite will be developed to configure and control the RF circuitry and AI enabled intelligence to process extracted data. This also provides real-time visualization of received signals, spectrum activity, radar data and system status.
3. **Conference or Journal paper**:A technical paper documenting the system architecture, design methodology, implementation challenges, and experimental results will be prepared for submission to a recognized conference or journal.
4. **Mandatory artifacts**: The project will produce all required academic deliverables, including a comprehensive final report, project presentation, demonstration video, and supporting documentation.

4. Resource Requirements

- RF simulation tools (ADS, AWR, CST, HFSS, or equivalent)
- High-frequency PCB fabrication and assembly
- Broadband Vivaldi antennas and patch antennas
- RF LNAs, mixers, switches, tunable filters, and IF amplifiers
- Embedded processing platform (STM32, Zynq, Raspberry Pi, or equivalent)
- Spectrum analyzer, VNA, signal generator, and RF measurement equipment
- FPGA/DSP resources for signal processing
- MATLAB/Python-based signal processing and AI development environment

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